Inner portion of the cylinder is called as tube and the gap between the outer portion of the inside tube and that of the inner portion of the outside tube is used for fluid storage and it is called as reservoir. A gland is fixed at the top portion of the cylinder. When damper is in operation, Gland is used to scrap the excess oil in the piston rods to the reservoir. When suspension works, fluid from the glands has been sent it to reservoir through piston rods.

The lower portion below the piston of the inner cylinder, is called as compression chamber and the top portion above the piston is called as rebound chamber. Damping fluid is made up of with 60% transform oil blended with 40% turbine oil.

Working principle

When the vehicle passes over the road disturbance, it piston moves down and this movement of piston pressurizes the fluid at the bottom portion. Then this fluid flows into the rebound chamber from compression chamber through a valve located on the top portion of the piston and the remaining portion flows into the reservoir through foot valves.

Similarly, when the vehicle comes to normal road condition, the piston moves from downward to the upward direction, so that the fluid at the top portion of the piston is compressed. This pressurized fluid is sent inside the compression chamber through the piston valve and the remaining amount of the fluid which again fed back from the reservoir through the foot valves. Because of this upward and downward motion of the piston, the fluid becomes highly pressurized and hence the energy due to road shock has been absorbed.

7.5.3 Air type

Air suspension is nowadays the most popular on commercial vehicles. It works on the principle of air so it is called as air type shock absorber.

7.6 Troubles, Causes and remedies of the suspension system

7.6.1 Noise in the suspension system

Sl. No.	Causes	Remedies
1	Loose, worn, un lubricated spring or suspension parts	Lubricate, tighten or repair
2	Worn out bushes	Replace the bush
3	Weak / Broken / Defective spring	Replace
4	Worn out shackle pin	Replace

7.6.2 Frequent failure in leaf spring

Sl. No.	Causes	Remedies
1	Defective shock absorber	Replace
2	Vehicle heavily loaded	Avoid overload
3	Tight spring shackle	Loosen, Rectify / Replace
4	Sudden brake in high speed	Apply brake gradually

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